

Chapter 27: Birds

Origin and Relationships

- **Archaeopteryx lithographica**: The earliest known bird fossil (147 million years old).
 - Demonstrated phylogenetic relatedness of birds and **theropod dinosaurs**.
 - Had reptilian skeleton (long bony tail, clawed fingers, abdominal ribs) but possessed **feathers**.



Figure 1: Archaeopteryx, a 147-million-year-old ancestor of modern birds. A, Cast of the second and most nearly perfect fossil of Archaeopteryx, which was discovered in a Bavarian stone quarry. Ten specimens of Archaeopteryx have been discovered, the most recent one described in 2007. B, Reconstruction of Archaeopteryx.

- **Evolutionary History:**
 - Birds are archosaurs, sharing derived characters with dromeosaurs (e.g., **Velociraptor**): furcula (fused clavicles), lunate wrist bones.
 - Living birds (**Neornithes**) are divided into:
 - **Paleognathae**: Flightless ostrich-like birds (ratites) and tinamous. Flat sternum (except tinamous).
 - **Neognathae**: All other birds, mostly flying, with a keeled sternum.



Figure 2: Evolution of modern birds. Of 30 living bird orders, 9 of the more important are shown. The earliest known bird, Archaeopteryx, lived in the Upper Jurassic, about 147 million years ago. Archaeopteryx uniquely shares many specialized aspects of its skeleton with the smaller theropod dinosaurs and is considered to have evolved within the theropod lineage. Evolution of modern bird orders occurred rapidly during the Cretaceous and early Tertiary periods.



Figure 3: Cladogram of the Saurischia, showing the relationship of several taxa to modern birds. Shown are a few of the shared derived characters, mostly related to flight, that were used to construct the genealogy. The ornithischians are the sister group to the saurischians and all are members of the clade Archosauria.

- **Flightlessness:** Evolved independently in many groups (e.g., penguins, ostriches), often on islands lacking predators.
 - Associated with loss of keel on sternum and heavy flight muscles.



Figure 4: One of the strangest birds in a strange land, a flightless cormorant, **Nannopterum harrisi**, of the Galápagos Islands dries its wings after a fishing forage. It is a superb swimmer, propelling itself through the water with its feet to catch fish and octopuses. The flightless cormorant is an example of a carinate bird (having a keeled sternum) that has lost the keel and the ability to fly.

Structural and Functional Adaptations for Flight

Feathers

- Unique feature of birds; lightweight, tough, and tensile.
- **Structure:** Hollow quill (**calamus**), shaft (**rachis**), and **barbs** forming a **vane**. Barbs have **barbules** with hooks that zip together.
- **Types:**
 - **Contour feathers:** Give outward form and flight.
 - **Down feathers:** Insulation (lack hooks).
 - **Filoplumes:** Hairlike, degenerate feathers.
 - **Powder-down:** Release waterproofing powder.
- **Molting:** Gradual replacement of feathers (usually once a year).



Figure 5: Types of bird feathers and their development. A to E, Successive stages in development of a vaned, or contour, feather. Growth occurs within a protective sheath, D, that splits open when growth is complete, allowing the mature feather to spread flat. F to H, Other feather varieties, including a pheasant feather with aftershaft, F, filoplumes, G, and down feathers, H.



Figure 6: Osprey, **Pandion haliaetus** (order Falconiformes), lands while holding a freshly captured fish. Feathers are molted in sequence in exact pairs so that balance is maintained during flight.

Skeleton

- **Pneumatized bones:** Light, delicate, laced with air cavities, yet strong.
- **Skull:** Diapsid (modified), fused into one piece, large braincase and orbits. Toothless keratinized beak.
- **Rigidity:** Fused vertebrae (**synsacrum**, **pygostyle**), ribs with **uncinate processes**.

- **Sternum:** Bears a large **keel** for flight muscle attachment.
- **Furcula:** Fused clavicles (wishbone), stores energy during wing beats.



Figure 7: A, Skeleton of crow showing portions of the flight feathers. B, Skeleton of Archaeopteryx showing reptilian structures (blue color) that are retained, modified, or lost in modern birds. The furcula and lunate wristbone (red color) were structures new to birds and their dromeosaur ancestors.

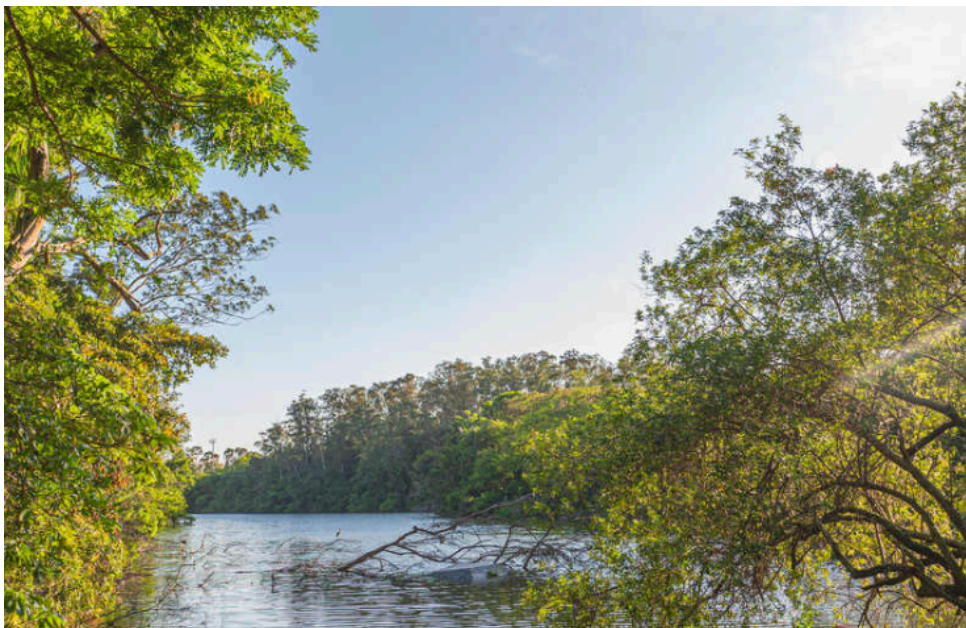


Figure 8: Hollow wing bone of a songbird showing stiffening struts and air spaces that replace bone marrow. Such “pneumatized” bones are remarkably light and strong.

Muscular System

- **Pectoralis:** Depresses wing (power stroke).
- **Supracoracoideus:** Raises wing; positioned on breast (low center of gravity), pulls via tendon over shoulder (“rope-and-pulley”).

- **Leg Muscles:** Concentrated in thigh; tendons extend to toes (prevents freezing, lightweight feet).
- **Toe-locking mechanism:** Automatically grips perch when settling.



Figure 9: Flight muscles of a bird are arranged to keep the center of gravity low in the body. Both major flight muscles are anchored on the sternum keel. Contraction of the pectoralis muscle pulls the wing downward. Then, as the pectoralis relaxes, the supracoracoideus muscle contracts and, acting as a pulley system, pulls the wing upward.



Figure 10: Perching mechanism of a bird. When a bird settles on a branch, tendons automatically tighten, closing the toes around the perch.

Food, Feeding, and Digestion

- **High Metabolic Rate:** Requires large amounts of food. Small birds eat more relative to body mass.

- **Beaks:** Adapted to food habits (e.g., generalized, seed cracking, probing, straining, tearing meat).



Figure 11: Some bills of birds showing variety of adaptations.

- **Digestive System:**
 - **Crop:** Storage; produces “crop milk” in pigeons.
 - **Stomach:**
 - **Proventriculus:** Secretes gastric juice.
 - **Gizzard:** Grinds food (keratinized plates, grit).
 - **Ceca:** Fermentation chambers in herbivores.
 - **Cloaca:** Common exit for digestive, excretory, and reproductive systems.

Circulatory and Respiratory Systems

- **Circulatory System:**
 - 4-chambered heart (complete separation of pulmonary/systemic).
 - Right aortic arch persists (unlike left in mammals).
 - Nucleated erythrocytes.
- **Respiratory System:**
 - **Parabronchi:** Tubelike lungs for continuous air flow.
 - **Air Sacs:** 9 interconnecting sacs; allow one-way flow of air through lungs (takes 2 cycles for one breath).
 - Highly efficient for high altitude and metabolic demands; also cools the body.



Figure 12: Respiratory system of a bird. A, Lungs and air sacs. One side of the bilateral air sac system is shown. B, Movement of a single volume of air through a bird's respiratory system. Two full respiratory cycles are required to move air through the system.

Excretory System

- **Kidneys:** Metanephric; excrete **uric acid** (semisolid, water conservation).
- **Salt Glands:** Located above eyes in marine birds; excrete excess salt (NaCl).



Figure 13: Salt glands of a marine bird (gull). One salt gland is located above each eye. Each gland consists of several lobes arranged in parallel. One lobe is shown in cross section, much enlarged. Salt is secreted into many radially arranged tubules, then flows into a central canal that leads into the nose.

Nervous and Sensory Systems

- **Brain:** Well-developed cerebellum (coordination) and optic lobes (vision). Enlarged dorsal ventricular ridge in cerebrum (instinctive behavior).



Figure 14: Bird brain showing principal divisions.

- **Senses:**
 - **Vision:** Keenest in animal kingdom. Large eyes, **pecten** (nutrients), **foveae** (sharp focus). Some see UV light.
 - **Hearing:** Good sensitivity; cochlea shorter than mammals.



Figure 15: A hawk eye has all the structural components of a mammalian eye, plus a peculiar pleated structure, the pecten, believed to provide nourishment to the retina. The extraordinarily keen vision of hawks is attributed to the extreme density of cone cells in the foveae: 1.5 million per fovea compared to 0.2 million for humans.

Flight

- **Aerodynamics:**

- Wing is an airfoil (cambered).
- **Lift:** Generated by negative pressure on upper surface and positive pressure below.
- **Slots:** **Alula** and gaps between primaries prevent stalling at low speeds/high angles of attack.



Figure 16: Air patterns formed by an airfoil, or wing, moving from right to left. At low speed the angle of attack must increase to maintain lift but this increases the threat of stalling. The upper figures show how low-speed stalling can be prevented with wing slots. Wing tip vortex (bottom), a turbulence that tends to develop at high speeds, reduces flight efficiency. The effect is reduced in wings that sweep back and taper to a tip.

- **Flapping Flight:** Primaries provide thrust; secondaries provide lift.



Figure 17: In normal flapping flight of strong fliers like ducks, the wings sweep downward and forward fully extended. Thrust is provided by the primary feathers at the wing tips. To begin the upbeat, the wing is bent, bringing it upward and backward. The wing then extends, ready for the next downbeat.

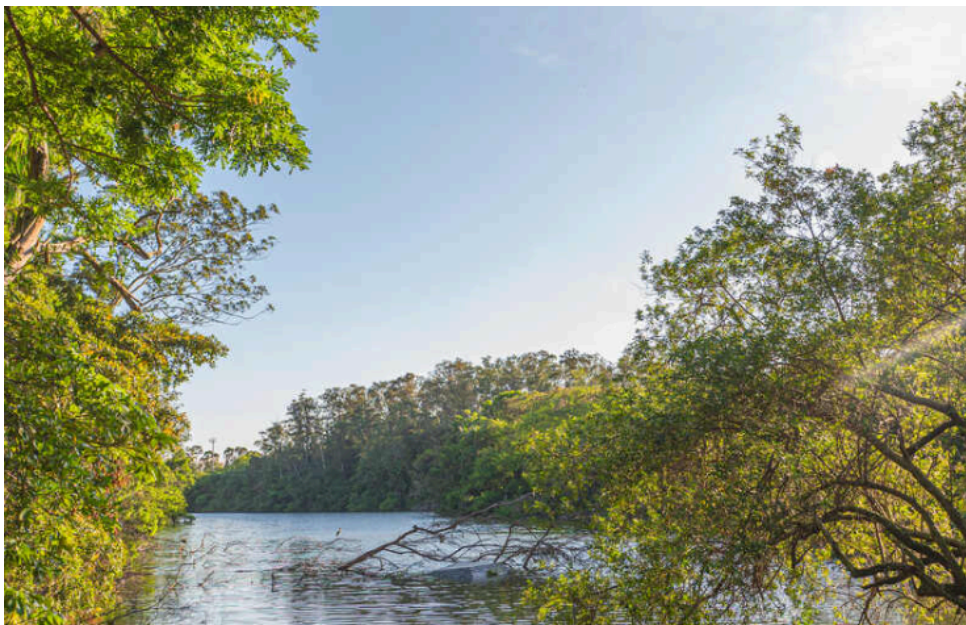


Figure 18: The secret of a hummingbird's ability to change direction instantly, or hang motionless in the air while sipping nectar from a flower, lies in its wing structure. The wing is nearly rigid, but hinged at the shoulder by a swivel joint and powered by a supracoracoideus muscle that is unusually large for the bird's size. When hovering the wing moves in a sculling motion. The leading edge of the wing moves forward on the forward stroke, then swivels nearly 180 degrees at the shoulder to move backward on the backstroke. The effect is to provide lift without propulsion on both forward and backward strokes.

- **Wing Types:**

- **Elliptical:** Maneuverability (sparrows, doves). Low aspect ratio, slotted.

- **High Aspect Ratio:** High speed (swallows) or dynamic soaring (albatrosses). Long, narrow, no slots.
- **High-Lift:** Static soaring, carrying heavy loads (hawks, eagles). Broad, slotted wings.



Figure 19: Four basic forms of bird wings.

Migration and Navigation

- **Migration:** Seasonal movement between breeding and wintering grounds.
 - Stimulus: Lengthening days (photoperiod) stimulate pituitary and gonads.
- **Navigation:**
 - Visual landmarks.
 - **Sun-azimuth orientation** (requires internal clock).
 - **Star patterns** (North Star axis).
 - **Magnetic field** (magnetite in head/neck).



Figure 20: Migrations of bobolinks, ***Dolichonyx oryzivorus***, and American golden plovers, ***Pluvialis dominica***. Bobolinks commute 22,500 km (14,000 miles) each year between nesting sites in North America and their range in Argentina, where they spend the northern winters, a phenomenal feat for such a small bird. Although the breeding range has extended to colonies in western areas, these birds take no shortcuts but adhere to the ancestral seaboard route. American golden plovers fly a loop migration, striking out across the Atlantic in their southward autumnal migration but returning in the spring by way of Central America and the Mississippi Valley because ecological conditions are more favorable at that time.

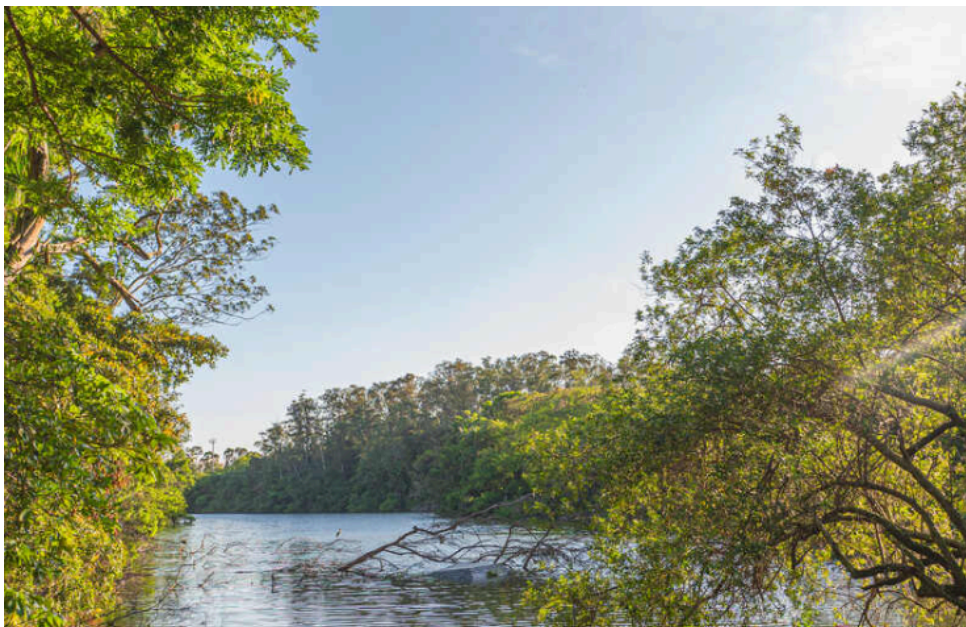


Figure 21: Gustav Kramer's experiments with sun-compass navigation in starlings. A, In a windowed, circular cage, the bird fluttered to align itself in the direction it would normally follow if it were free. B, When the true angle of the sun is deflected with a mirror, the bird maintains the same relative position to the sun. This shows that these birds use the sun as a compass. The bird navigates correctly throughout the day, changing its orientation to the sun as the sun moves across the sky.

Social Behavior and Reproduction

- **Sociality:** Flocking for protection, migration, or feeding (e.g., pelicans).



Figure 22: Part of a colony of northern gannets, **Morus bassanus**, showing extremely close spacing between pairs in this highly social bird. Order Pelecaniformes.

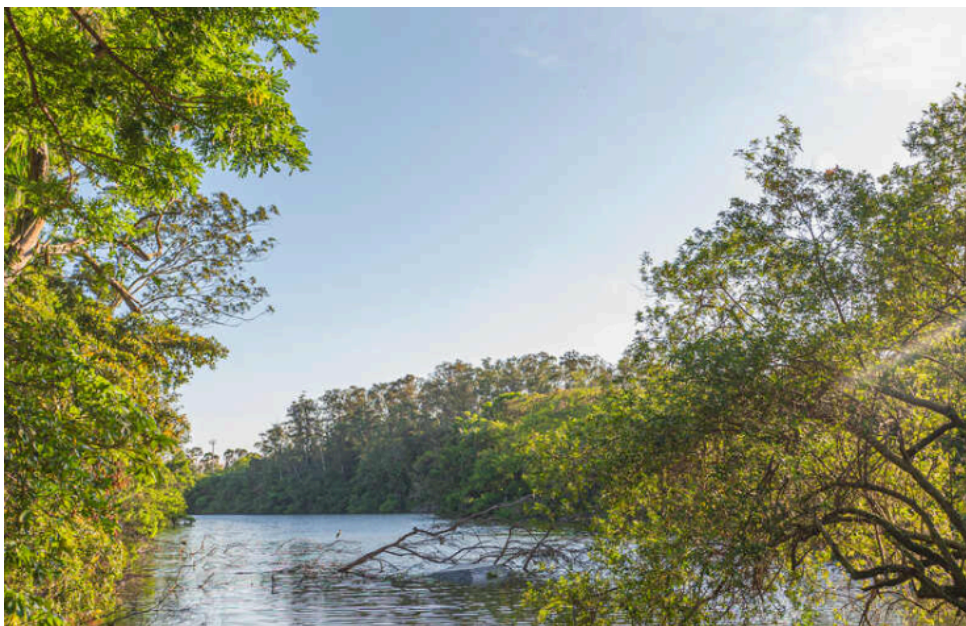


Figure 23: Cooperative feeding behavior by white pelicans, **Pelecanus onocrotalus**. A, Pelicans form a horseshoe to drive fish together. B, Then they plunge simultaneously to scoop fish in their huge bills. These photographs were taken 2 seconds apart.

- **Reproduction:**
 - Testes enlarge seasonally.
 - Females: Only left ovary/oviduct functional.
 - Fertilization in upper oviduct; albumin and shell added later.



Figure 24: Copulation in waved albatrosses, ***Diomedea irrorata***. In most bird species males lack a penis. A male copulates by standing on the back of a female, pressing his cloaca against that of the female, and passing sperm to the female.

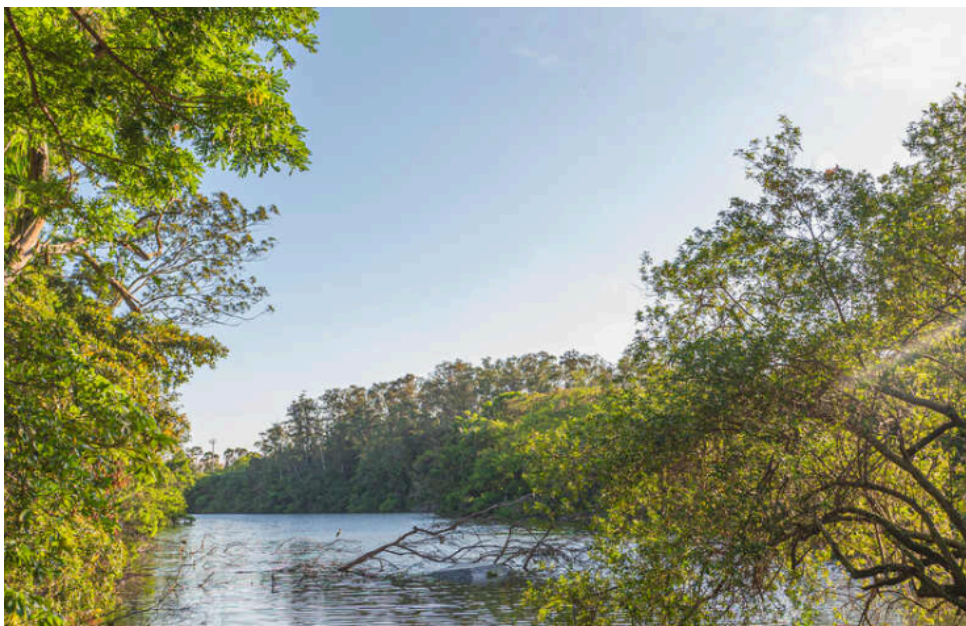


Figure 25: Reproductive system of a female bird. For most birds, only the left ovary and reproductive tract are functional. Structures on the right dwindle to vestiges.

- **Mating Systems:**

- **Monogamy:** Common (>90%). Both parents care for young.
- **Polygyny:** One male, many females (e.g., grouse in **leks**).
- **Polyandry:** One female, many males (rare, e.g., spotted sandpipers).



Figure 26: Dominant male greater sage grouse, **Centrocercus urophasianus**, surrounded by several hens that have been attracted by his “booming” display.

- **Nesting and Young:**

- **Altricial:** Naked, helpless at birth (e.g., songbirds).
- **Precocial:** Down-covered, active at birth (e.g., ducks, quail).



Figure 27: Anna’s hummingbird, **Calypte anna**, feeding its young in its nest of fibers and down from plants, bound together with spiderwebs and camouflaged with lichens. The female builds the nest, incubates two pea-sized eggs, and rears the young with no assistance from a male. Anna’s hummingbird is a common resident of California. It is the only hummingbird to overwinter in the United States.



Figure 28: Comparison of one-day-old altricial and precocial young. The altricial meadowlark (left) hatches nearly naked, blind, and helpless. The precocial ruffed grouse (right) is covered with down, alert, strong legged, and able to feed itself.

Bird Populations

- Populations vary with food supply.
- Human impact: Introductions (starlings), extinctions (passenger pigeon), habitat destruction.

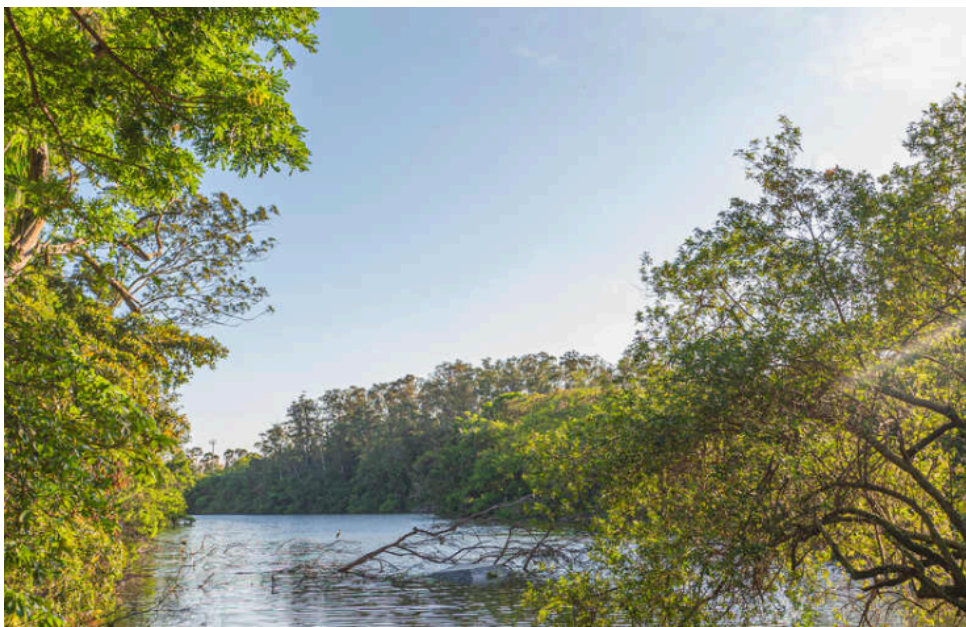


Figure 29: Colonization of North America by starlings, ***Sturnus vulgaris***, after the introduction of 120 birds into Central Park in New York City in 1890. There are now perhaps 200 million starlings in the United States alone, testimony to the great reproductive potential of birds. Starlings are omnivorous, eating mostly insects in spring and summer and shifting to wild fruits in the fall.



Figure 30: Sport-shooting passenger pigeons in Louisiana during the nineteenth century. Relentless sport and market hunting, before establishment of state and federal hunting regulations, in addition to clearing of the hardwood forests that served as nesting habitats, eventually dropped the population too low to sustain colonial breeding. The last passenger pigeon died in captivity in 1914.

Classification Examples

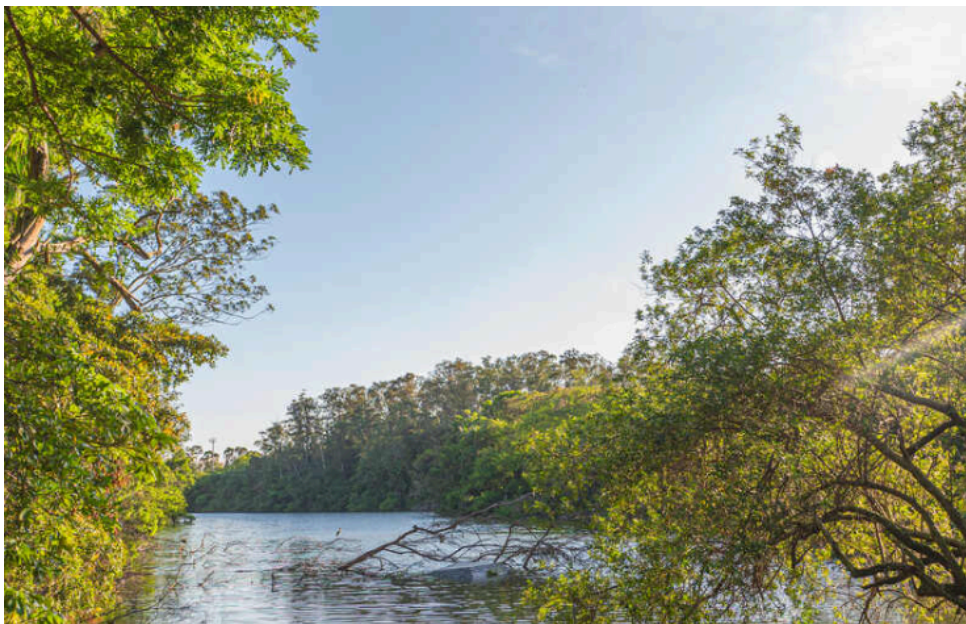


Figure 31: Ostrich, **Struthio camelus**, of Africa, the largest of all living birds. Order Struthioniformes.



Figure 32: Greater flamingos, **Phoenicopterus ruber**, on an alkaline lake in East Africa.
Order Phoenicopteriformes.



Figure 33: Laughing gulls, **Larus atricilla**, in flight. Order Charadriiformes.



Figure 34: Ground finch, **Geospiza fuliginosa**, one of the famous Darwin's finches of the Galápagos Islands. Order Passeriformes.